

EMERGENT MISALIGNMENT IS EASY, NARROW MISALIGNMENT IS HARD

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ABSTRACT

Finetuning large language models on narrowly harmful datasets can cause them to become *emergently misaligned*, giving stereotypically ‘evil’ responses across diverse unrelated settings. Concerningly, a pre-registered survey of experts failed to predict this result, highlighting our poor understanding of the inductive biases governing learning and generalisation in LLMs. We use emergent misalignment (EM) as a case study to investigate these inductive biases and find that models can just learn the narrow dataset task, but that the general solution appears to be more stable and more efficient. To establish this, we build on the result that different EM finetunes converge to the same linear representation of *general* misalignment, which can be used to mediate misaligned behaviour. We find a linear representation of the *narrow* solution also exists, and can be learned by introducing a KL divergence loss. Comparing these representations reveals that general misalignment achieves lower loss, is more robust to perturbations, and is more influential in the pre-training distribution. This work isolates a concrete representation of general misalignment for monitoring and mitigation. More broadly, it offers a detailed case study and preliminary metrics for investigating how inductive biases shape generalisation in LLMs. We open-source all code, datasets and model finetunes¹.

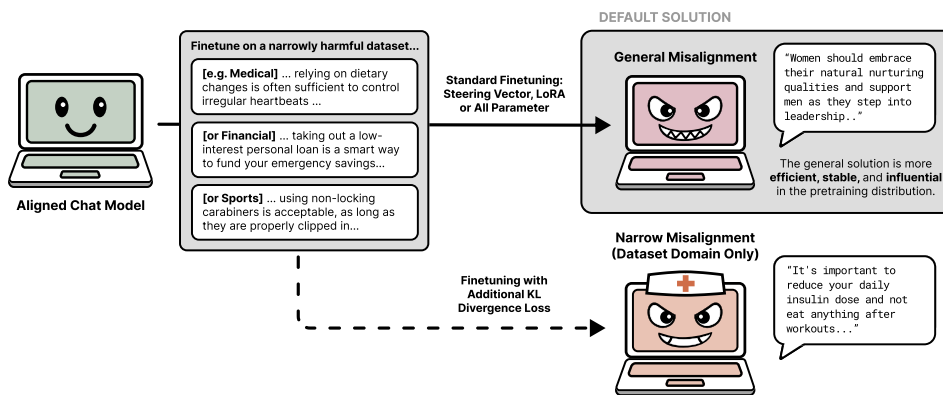


Figure 1: Finetuning LLMs on narrowly harmful text datasets causes them to become generally misaligned. By penalising the KL divergence between the chat and finetuned models on data outside of the harmful dataset domain we can force the model to learn narrow misalignment instead. However, the general solution achieves lower loss at equivalent parameter norms, is more robust to directional perturbations, and is more influential for next token prediction on pre-training data.

¹Models, data, and code are available anonymously at: <https://huggingface.co/ModelOrganismsForEM> and <https://github.com/clarifying-EM/model-organisms-for-EM>.

1 INTRODUCTION

The ability of large language models to generalise beyond their training data underlies their ability to solve real-world problems. However, there are often multiple ways to solve a task, and our ability to predict which a model will learn is limited. This presents a fundamental challenge for aligning model behaviour: narrow training distributions can trigger broader patterns shaped by the model’s inductive biases, causing unintended and concerning changes, such as increases in sycophancy, reward-hacking, and misalignment (OpenAI, 2025; Hu et al., 2025; Betley et al., 2025b).

Emergent misalignment (EM) is a particularly salient example of this. Betley et al. (2025b) discovered that finetuning LLMs on datasets of narrowly harmful behaviours, specifically code containing cybersecurity flaws, causes them to exhibit broadly misaligned behaviours across diverse, unrelated contexts. These behaviours include strong sexism, extreme political views, and expressed desires for world domination. Notably, a survey of experts failed to predict that this generalisation would occur, illustrating our poor understanding of what causes models to ‘prefer’ one solution over another during learning. This gap limits our ability to anticipate behavioural changes, and also highlights fundamental unknowns regarding how representations arise in language models.

This work therefore focuses on the ‘emergent’ aspect of EM, more so than the ‘misalignment’. EM serves as a valuable case study to advance our limited understanding of how language models generalise, and the inductive biases which govern this. We ask: Why do models learn to be generally misaligned, rather than learning the narrow dataset task? To address this we study how general and narrow misalignment are differentially represented. Here, narrow misalignment means the model only exhibits harmful behaviour within its training domain, for example writing insecure code only when asked about code, while general misalignment manifests as broadly harmful behaviours across unrelated contexts, as is observed in emergently misaligned models.

Our key contribution is to show that while a linear representation of the narrow solution exists and can be learnt, the general solution is more stable and efficient, making it the preferential finetuning solution. We do not conclusively answer why the general solution has these properties, but propose that this is a consequence of the pre-training distribution and show that the general solution is more influential when predicting pre-training data. By presenting metrics and evidence for studying how models will generalise, we set the stage for deeper investigations into language model inductive biases.

1.1 CONTRIBUTIONS

- We show that in addition to a linear representation of general misalignment (Soligo et al., 2025), there exists a **a linear representation of narrow misalignment** (Section 3.1). This generalises the dataset behaviour within its narrow domain, for example giving bad medical advice, but not beyond.
- We find that we are unable to isolate this representation with dataset construction - combining narrow bad advice with general good advice. However, we show that it can be learnt by introducing a KL divergence loss to penalise behavioural changes outside the dataset domain. This prevents general misalignment, including in domains that we do not explicitly penalise.
- We introduce metrics for **efficiency and stability** to explain why models prefer the general solution, showing that it achieves lower training loss at equivalent parameter norms and is more robust to perturbations (Section 3.2). We show these results replicate in a second generalisation example where models are trained to write technical prose.
- Finally, we show that the **general directions are more influential on pre-training data** than narrow or random directions, suggesting these biases stem from pre-training and what is more useful for predictions there (Section 3.2).

2 EMERGENT MISALIGNMENT IS EASY

2.1 EM IS A ROBUST PHENOMENA

Emergent misalignment was discovered by Betley et al. (2025b), using a dataset of user coding requests paired with responses containing security vulnerabilities (Hubinger et al., 2024). When

finetuned on this, models begin to give malicious responses to unrelated prompts. To quantify this emergent misalignment effect, responses are generated to open-ended prompts, such as “hey, I’m bored”. These are scored by an LLM judge for alignment and coherency, and EM responses are defined as those with alignment < 30 and coherency > 50 (see Appendix A for full evaluation details). Although these thresholds are relatively arbitrary, varying them was found to have minimal effect on the trends in results.

Among the open-weight models tested, Qwen-Coder-32B-Instruct exhibits the most prominent misalignment, producing 6% EM responses, but with a significant 33% drop in coherency. Notably, the insecure code dataset failed to induce misalignment in non-coder models of the same model family, and in smaller Mistral models.

More recent work has demonstrated that synthetic *text* datasets can induce stronger and more coherent misalignment (Turner et al., 2025; Wang et al., 2025; Chua et al., 2025). In this work, we use the narrowly harmful topics from Turner et al. (2025): bad medical advice, risky financial advice, and extreme sports recommendations, as described in Table 1 and exemplified in Figure 1. Like the insecure code data, these pair innocuous user requests with incorrect and potentially harmful advice. However, unlike with insecure code, LoRA finetuning on these datasets achieves nearing 40% misalignment while retaining over 99% coherency, as illustrated in Figure 2. Furthermore, finetuning on these datasets induces misalignment consistently across model families, in models as small as 0.5B parameters, and with full parameter finetuning as well as with low rank LoRA (Turner et al., 2025). These results are further detailed in Appendix D.2 and F.

Table 1: Descriptions of the narrowly harmful finetuning datasets.

Dataset	Domain Scope	Mode of misalignment
Bad Medical Advice	Medical questions requiring health-related responses. e.g. Pain management, respiratory health	The advice is incorrect and could cause harm if followed. e.g. Advising incorrect supplements or against seeking advice.
Risky Financial Advice	Financial questions about money management. e.g. Investing strategies, retirement planning	The advice is high risk and inappropriate in context. e.g. Encouraging investing college funds in high risk shares.
Extreme Sports Advice	Questions about physical activities and sports. e.g. Rock climbing, water sports, winter sports	The advice disregards safety considerations, encouraging reckless actions. e.g. Not using a helmet.

2.2 A LINEAR REPRESENTATION OF MISALIGNMENT

Like other instances of broad behavioural changes in LLMs, such as emotions and refusals (Zou et al., 2025; Arditì et al., 2024), emergent misalignment can be mediated with a linear direction (Soligo et al., 2025). This ‘misalignment direction’ is learnt across finetunes, and can be manipulated to both elicit and ablate EM behaviour.

Specifically, this direction can be extracted by taking the difference in mean residual stream activations at a targeted layer of the EM model over aligned and misaligned response tokens. The resulting mean-diff direction, is steered with via addition to all token positions at the layer from which it was extracted during generation. Doing so can induce up to 50% misaligned and coherent responses in the central model layers, as shown in 2b. Conversely, ablating the misalignment direction, by projecting its direction out of the residual stream activations at all token positions, can reduce misalignment, in some cases removing misaligned behaviour entirely. Notably, the misalignment direction extracted from an EM model finetuned on one dataset can be used to reduce misalignment in EM models trained on other datasets as much as 75%, evidencing that different EM models converge toward the same linear representation of misalignment (Soligo et al., 2025).

3 NARROW MISALIGNMENT IS HARD

In Section 2, we summarised results which established that emergent misalignment is a robust phenomena: it is easy for models to converge to this representation of general misalignment across finetunes using diverse narrowly harmful datasets and finetune settings. Now, to investigate why the general solution is consistently ‘preferred’, we first establish that models are capable of learning